OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 35 OF 55

REAL-TIME DRIVER DROWSINESS DETECTION USING FACIAL LANDMARK ANALYSIS AND DEEP LEARNING

Domain: Automotive Safety & In-Cabin Telematics | **Module:** project_35_driver_drowsiness

OFFICIAL ABSTRACT

Driver drowsiness is one of the leading causes of traffic accidents worldwide, resulting in significant loss of life and property damage. Developing an automated system that monitors driver attentiveness in real-time is essential to mitigate these risks and enhance road safety. This study presents a non-invasive, real-time driver drowsiness detection system that utilizes facial landmark analysis combined with deep learning techniques. The system uses a standard camera positioned on the vehicle dashboard to capture the driver's facial features under varying lighting conditions. Facial landmark analysis is employed to locate key facial regions, specifically the eyes, mouth, and head pose. Key metrics such as the Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) are calculated to track eye closure duration and yawning frequency. A deep learning model, specifically a Long Short-Term Memory (LSTM) network, is then used to analyze temporal patterns in these metrics to distinguish between normal driving behaviors and signs of fatigue or distraction. The model is trained on a diverse dataset of facial expressions and simulated drowsiness scenarios, achieving robust generalization across different individuals. Experimental results show high accuracy in detecting early stages of drowsiness with minimal false alarms. Integrating this system into vehicles provides a reliable early-warning mechanism that alerts drivers using auditory and visual prompts, thereby significantly reducing fatigue-related accidents and improving driver safety.

📌 Project Summary & System Specifications

Project Title	REAL-TIME DRIVER DROWSINESS DETECTION USING FACIAL LANDMARK ANALYSIS AND DEEP LEARNING
Specialization / Category	Automotive Safety & In-Cabin Telematics

Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5035`
Local Implementation	`project_35_driver_drowsiness`